

Problem

Let $\mathbf{X} = 4\angle 40^\circ$ and $\mathbf{Y} = 20\angle -30^\circ$. Evaluate the following quantities and express your results in polar form:

(a) $(\mathbf{X} + \mathbf{Y})\mathbf{X}^*$

(b) $(\mathbf{X} - \mathbf{Y})^*$

(c) $(\mathbf{X} + \mathbf{Y})\mathbf{X}$

Step-by-step solution

Step 1 of 3

(a)

Consider the function, $f = (\mathbf{X} + \mathbf{Y})\mathbf{X}^*$.

Substitute $4\angle 40^\circ$ for $\mathbf{X}$ and $20\angle -30^\circ$ for $\mathbf{Y}$ to obtain $(\mathbf{X} + \mathbf{Y})\mathbf{X}^*$.

$$\begin{aligned} f &= (\mathbf{X} + \mathbf{Y})\mathbf{X}^* \\ &= (4\angle 40^\circ + 20\angle -30^\circ)(4\angle 40^\circ)^* \\ &= (3.064 + j2.57 + 17.32 - j10)(3.064 + j2.57)^* \\ &= (20.384 - j7.43)(3.064 - j2.57) \end{aligned}$$

Simplify further.

$$\begin{aligned} f &= 43.36 - j75.15 \\ &= 86.76\angle -60.015^\circ \end{aligned}$$

Hence, the polar form f is $\boxed{86.76\angle -60.015^\circ}$.

Step 2 of 3

(b)

Consider the function, $f = (\mathbf{X} - \mathbf{Y})^*$.

Substitute $4\angle 40^\circ$ for $\mathbf{X}$ and $20\angle -30^\circ$ for $\mathbf{Y}$ to find $(\mathbf{X} - \mathbf{Y})^*$.

$$\begin{aligned} f &= (\mathbf{X} - \mathbf{Y})^* \\ &= (4\angle 40^\circ - 20\angle -30^\circ)^* \\ &= (3.064 + j2.57 - 17.32 + j10)^* \\ &= (-14.256 + j12.57)^* \end{aligned}$$

Simplify further.

$$\begin{aligned} f &= -14.256 - j12.57 \\ &= 19\angle -138.5^\circ \end{aligned}$$

Hence, the polar form f is $\boxed{19\angle -138.5^\circ}$.

Step 3 of 3

(c)

Consider the function, $f = \frac{(X+Y)}{X}$.

Substitute $4\angle 40^\circ$ for X and $20\angle -30^\circ$ for Y to find $(X+Y)X^*$.

$$\begin{aligned} f &= \frac{(X+Y)}{X} \\ &= \frac{(4\angle 40^\circ + 20\angle -30^\circ)}{(4\angle 40^\circ)} \\ &= \frac{(3.064 + j2.57 + 17.32 - j10)}{(3.064 + j2.57)} \\ &= \frac{(20.384 - j7.43)}{(3.064 + j2.57)} \end{aligned}$$

Simplify further.

$$\begin{aligned} f &= 2.711 - j4.699 \\ &= 5.425\angle -60^\circ \end{aligned}$$

Hence, the polar form f is $\boxed{5.425\angle -60^\circ}$.